

Philipps-Universität Marburg
Fachbereich Mathematik und Informatik

Bachelor's thesis

User Stories Quality Analyse Using Secure Generative AI

Dependency and Independency between User Stories

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1 Motivation

1.1 Agile Development and User Stories

How important is agile development and so that the importance of user stories? implies the quality of user stories can affect the quality of the software quality.

In requirement engineering, agile development is a widely adopted method to enhance the waterfall method of software development. Agile development enables rapid development: It allows for iterative, small-scale, and rapid development and functionality delivery, leading to a rapid response to user requirements [5]. In the artificial intelligence era, a software prototype can be deployed using agile development for requirement analysis in no time.

Agile Manifesto:The Agile Manifesto values (Individuals and interactions, Working software, Customer collaboration, Responding to change) are presented as better ways of developing software compared to the values of traditional approaches (processes and tools, comprehensive documentation, contract negotiation, following a plan).[5]

Scrum is the widely adopted agile framework[14]. Scrum contains roles, artifacts, and processes[5]. There are many layers of artifacts in SCRUM such as Epic, Feature, User Stories, and Product Backlog. These are textual high-level user requirements in SCRUM.

User stories, which are fundamental for the successful software development, are the starting base of agile frameworks[5]. Shifting-left of the quality check is needed to control software quality in early phases and to increase customer satisfaction.

1.2 Quality of User Stories

US needs to capture all essential elements of a user requirement. In order to evaluate the quality of USs, many researchers have already defined different quality frameworks to check the quality not only user requirements but also of a US and also a set of USs.

Leffingwell et al. defined a set of criteria called INVEST: Independent, Negotiable, Valuable to users or customers, Estimable, Small, and Testable.[7]

Lai defined three factors for evaluating US quality: Basic Quality (Discussable and Estimable), Management Quality (Controllable and Manageable), and Acceptance Quality (Confirmable). Each factor consists of three metrics, with different weights, and the final quality metric is the weighted sum of all three factors. They also suggested rules for correcting low-quality USs.[6]

Lucassen et al. presented the Quality User Story (QUS) framework, which is a collection of 14 rules that determine the quality of not only a US but also a set of USs.

Using this framework, they developed a tool called AQUSA, which uses NLP techniques to verify some of the QUS criteria automatically [11]. Each quality rule is classified according to three concepts from linguistics: syntactic, semantic, and pragmatic.The QUS framework and the AQUSA tool have been combined into a package called the Grimm Method [9].

Femmer et al. defined low-quality criteria of requirements as "requirement smells" [2], a concept inspired by Code Smells in software engineering. A prototype tool called *Smella* was developed to detect these quality issues automatically.

Xu et al. [15] introduced a user requirement quality model, which consists of 6 quality criteria for individual user requirements (Complete, Consistent, Simple, Accurate) and overall user requirements (Robust, Unique, Harmonious). User requirements are represented by USs.

1.3 Importance of User Story's Dependency

Why dependencies between user stories is important to tracing and marked?

1.4 Related work

1. A Systematic and Lightweight Method to Identify Dependencies between USs
2. User Story Quality Assessment Based on Multi-dimensional Perspective: A Preliminary Framework
3.

2 Fundamentals

2.1 User Story and its conceptual model

2.1.1 User Story

Leffinwell et al. defined in "A User Story Primer" a User Story as "a brief statement of intent that describes something the system needs to do for the user."[7]

Definition 2.1 — User story (US) is a user-centric natural language software requirement to describe a software feature. US is semi-structuredly notated when it follows a prescribed format (template). This template helps user stories captures key aspects: *who* wants the functionality, *what* functionality the end-users expect, and optionally why the end-users need this functionality.

There is no universal template of user stories but the mostly used is the Connextra template [10]: *As a <role>, I want <goal>,[so that <benefit>]"*. Or the other terminology is This kind of template defines the high-level representation of US. For further automatically quality analysis of USs, a more granular blueprint of US's structure is needed.

2.1.2 Conceptual Model

Hofmann et al. defined conceptual model like structured representation of user stories (USs) [4].

This definition is inspired by the informal definitions of Hofmann et al. and Arulmohan et al.

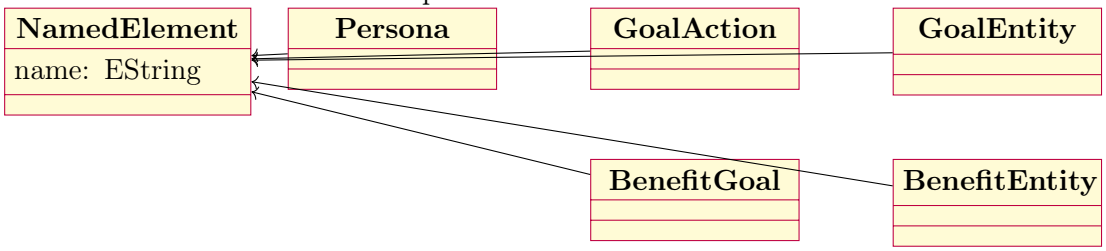
Definition 2.2 — A conceptual model of US captures US in conceptual schema that defines key components of a US and their relationships. The conceptual model can be illustrated by using diagrams such as UML diagrams (e.g. class diagram) in order to describe the relations between entities.

Since the researches of US started, there are some conceptual models derived from the Connextra template. Different level of abstraction can affected the conceptual model of a US.
Lucassen et al. had defined a conceptual models with 3 parts of a US, which are ROLE, MEANS, ENDS [12]. Each US has one ROLE, one MEANS, and zero to one ENDS. Each ROLE contains one indicator and one functional role. Each MEANS consists of one subject, one main object, one main verb, and zero to one free form. At last, the ENDS has the similar structure to the MEANS [12]. In other paper of the same authors,

they introduced that US contains 4 parts one role, one means, zero or more ends, and one format [11]. The second conceptual model is more detailed. This points out that conceptual model of USs can have different level of abstraction.
Arulmohan et al. also defined a metamodel for backlogs [1]. A *Backlog* contains one to N *US*, one to N *Persona*, and one to N *Entity*. A *US* consists of one *Persona*, one *Entity*, one *Action* , and zero to one *Benefit*. There are 2 one-side relations: a `trigger`-relation from *Persona* to *Action* and a `targets`-relation from *Action* to *Entity*.
Hofmann et al. adopted introduced metamodel of Arulmohan et al. to define the conceptual model for anotated USs [4]. The conceptual model structures user stories using annotation graphs, defining elements like *Persona, GoalAction, GoalEntity, BenefitAction, and BenefitEntity*. These components are connected by relationships, helping identify strict and semantic redundancies in agile development.
What is the advantage of using this conceptual model? Does the conceptual model depend on the dataset?

2.1.3 Inherited notation and quality definitions of user stories

This thesis is an extension of the paper of Hofmann et al.[4]. This bachelor thesis will focus on detecting dependency and independency pairwisely between USs. These are qualities of a set USs. The conceptual model of Hofmann et al. is used to to define the anotated USs, as well as the inherited defintions regarding "Annotated User Story" and "Well-formedness" of Hofmann et al. is used to define depedency types between USs. The inherited definitions and conceptual model is defined as follows:



Definition 2.3 (Annotated User Story) — [4] An annotated user story is a user story (US) together with an instance c of the conceptual model from Fig. 1 such that each *name*-attribute of a node of c contains a part of the US text; the instance c is then called the annotation graph of US.

We call a *targets*-edge together with incident *GoalAction* and *GoalEntity* an activity of the annotated user story and a *targets*-edge together with incident *BenefitAction* and *BenefitEntity* a reason.

Definition 2.4 (Well-formedness) — [4] An annotated user story is well-formed if it contains at least one activity and at most one reason

2.2 Semantic Knowledge

Define "sematic similarity", "semantic related", "semantic equivalent", "lexical identic"

2.3 Definition of Dependency and Independency

2.3.1 Independency

Requirement independency

There are not many papers regarding the dependencies between USs, so it is rational to firstly take a look into the requirement dependencies. On the requirement level, there are some papers which already automated the relation extraction using NLP and Machine Learning. Guan et al. [3] proposed an automatic approach to extract requirement dependency based on integrated active learning strategies. They defined "basis dependencies" and "advanced dependencies." Firstly, a pair of requirements can be checked with "basis dependencies", this means that this pair is either dependent or independent (XOR-relation). Then, if this pair is dependent, specific dependency types can be extracted, which are *Notification, Arouse, Call, Conflict, Aggregation, and Similar.* So they defined *independency* between **two** requirements as follow: Two requirements (R_x, R_y) are independent if no recognized dependency type (e.g., notification, arouse, call, conflict, aggregation, similar) exists between them; otherwise, they are considered dependent. The definition of 6 types of dependency will be cited in the below paragraph.

User Story independency

Some researchers has specifically contributed to this topic but only applied on the USs. Leffingwell et al. [7] defined *Independency* in "A User Story Primer" that a US can be developed, tested, and potentially delivered on its own, without being tied to other USs. This allows for more flexibility in prioritization and avoids dependencies that could slow down development. Ideally, each story should bring value on its own and be able to stand alone without requiring the completion of another story beforehand.
Lucassen et al. defined in their framework *Independency* based on the INVEST framework of Leffingwell et al. above. "User Stories should not overlap in concept and should be schedulable and implementable in any order".[11]
So based on the above researches, a definition of independency between two USs (pairwise) is developed.

Definition 2.5 (**Independency - Pairwise**) — Let U be a set of user stories, and a pair of USs US_i, US_j in U is called independent if there exists no type of dependency (operation-based, sequential, hierarchical, compositional dependency) between them, so that each of them can be implemented, tested for its fulfillment without affecting the other one.

For example, US_1 and US_2 are independent.
US_1: As a user, I want to change my profile picture.
US_2: As a user, I want to modify my address.

2.3.2 Dependencies

Requirement dependency types

Different papers with different terminology of dependency types in requirements engineering. For example:,

Guan et al. [3] developed a terminology that differentiates the six dependency types quote to quote as following: *Notification*, *Arouse*, *Call*, *Conflict*, *Aggregation*, *Similar*.

Motger et al. [13] introduced 6 relation types, which define how requirements depend on or affect each other. A *Requires* relation means one requirement must be fulfilled before another can proceed, while *Conflicts* shows that one requirement restricts another. If two requirements are mutually exclusive, they *Contradict* each other. *Is_a_variant* marks alternative ways to meet the same goal, and *Is_similar* highlights overlapping content or redundancy. Finally, *Details* means one requirement elaborates on another. These relations help structure requirements clearly, minimizing contradictions and overlaps to apply NLP methods.

Zang et al. [16] mentioned that various dependency terminologies has been proposed with different level of abstraction. They points out that there are some problems with existing dependency models that they have overlapping and vague types, some are hard to apply in practice and different users interpret them differently. These issues reduce clarity and usefulness, so the authors proposed a simpler, clearer model with better definitions. Dependencies are classified into 2 broad categories: intrinsic dependencies and additional dependencies. Intrinsic dependencies reflect the essential relationships that exist among requirements and directly impact various software engineering activities and most important they have clear affects at the implementation level, which are *Constrain:* This represents one requirement constraining another, often affecting implementation decisions or crosscutting concerns. *Precede:* This indicates a sequencing relationship where one function must precede another, influencing process flows or control structures in the code. *Be_similar_to:* This highlights requirements that share similar data or behavior, often leading to reuse or shared components in implementation. *Refine:* This reflects hierarchical relationships where a higher-level requirement is broken down into more detailed sub-requirements, influencing module or class design. *Be_exception_of:* This identifies exception handling scenarios, which are critical for defining error-handling paths in the code. *Conflict:* This shows where the implementation of one requirement may negatively impact another, requiring careful architectural decisions to resolve. On the other hand, additional dependencies represent relationships that are introduced by specific software engineering tasks such as product release planning or cost-value trade-off analysis.

Li et al.[8] researched and consolidated 23 dependency types into a dependency model, which contains 3 main groups. `Document-related dependencies` arise from the rela-

tionships and structure within requirement documents. `Value-related dependencies` indicate how the fulfillment of one requirement influences the perceived value or cost of another, helping prioritize requirements in release planning. `Implementation-related dependencies` focus on how requirements are technically realized and how their implementations depend on each other, particularly relevant for change impact analysis during software design and development phases. The authors decided to narrow the scope to implementation-related dependencies because they wanted to focus on the most impactful and traceable relationships, which are either revealed by BPMN. Five implementation-related dependency types are defined. The first, called *Constraints*, exits when a requirement can limit or restrict another.The second,*Refines*, is when a requirements adds more detail to another.The third type, *Precondition*, means something needs to happen first, such as a task being completed or a condition being met, before something else can proceed. This often relfects the order of steps in a process. The fourth type, *Satisfies*, shows that fulfilling one requirement also meets another. Finally, *Similar_to* is used when a requirement is very much like, or overlaps with, one or more others.

What we learned to apply to better define user stories dependencies?

User Stories dependency types

Leffingwell et al. mentioned that many USs have some "natural, sequential dependencies as the product functionality builds, and yet each piece can deliver value independently." [7]

Lucassen et al.[11] emphasized that some dependencies between USs are "practically impossible" to resolve because, analogous to the coupled system, some USs are linked to each other. They recommended establishing **explicit** dependencies, which can be solved by using issue tracker tools such as Jira and Azure DevOps. The dependencies of a US should be included as a link in the issue tracker tool. They also defined 2 types of US dependency relation between the two USs : *Causality* and *Superclasses*. A US_a is causally depended on US_b, if the US_a is "necessary" for starting development of the US_b. And a US_c is a *superclasses* of the US_d, when US_c is parent of US_d, which also means that US_c contains US_d.

For dependency management, some common agile project management tools e.g., Jira, Azure DevOps already integrated some default dependency types (linking). Such as in Jira, there are 4 default types *relates to, duplicates, blocks, clones* [1]. Azure DevOps also uses 5 common link types: *Affects, Duplicate, Parent/Child, Predecessor/Successor, Related*[2]

Only concentrate on functional User Stories, uses notion of Hofmann to concrete define. Inspiration of xxx

Definition 2.6 (**Operation-based Dependency (OBD) - Pairwise**) — Let U be a set of user stories, and let $US_i, US_j \in U$.

[1] *Configuring issue linking*, 27.06.2025. Jira documentation
[2] *Reference guide for link types*, 27.06.2025. Azure DevOps Learn

Let $A(US_i)$ and $A(US_j)$ be the sets of activities associated with US_i and US_j, respectively, where each activity $a \in A(US_i)$ (and $b \in A(US_j)$) is defined as a pair:

$$a = (a.\text{Action.name},\ a.\text{Entity.name}) \qquad b = (b.\text{Action.name},\ b.\text{Entity.name})$$

Specifically, *Action* denotes a *GoalAction* and *Entity* denotes a *GoalEntity*.
Define the following:

- requires(a.Action.name, b.Action.name): One operation (e.g., delete, view, share) described in Action.name can only be implemented or executed if other operation (e.g., create, add) has been performed. This expresses a strict dependency where the prerequisite operation provides the necessary existence or state of the entity required for subsequent actions.
- a.Entity.name $\approx$ b.Entity.name: Lexical identity or semantic equivalence between `name`-attributes of Entity of a and b.

An *Operation-based Dependency* exists between US_i and US_j **if and only if**

$$\exists\,(a,b) \in A(US_i) \times A(US_j) :\ a.\text{Entity.name} \approx b.\text{Entity.name} \\ \wedge\ \text{requires}\,\big(a.\text{Action.name},\ b.\text{Action.name}\big)$$

In words: An *operation-based dependency* between two user stories exists when an operation described in activity (GoalAction) in one story, such as a CRUD operation like read, delete, update or another operation like share, view, validate or export, can only be carried out if the operation described in activity (GoalAction) in the other story (e.g., create or add) has already been performed on the identical or a semantically equivalent entity (GoalEntity). This means the one user story relies on the other one to ensure the necessary existence or state of the entity before its own action can proceed.

Definition 2.7 (**Sequential Dependency - Pairwise**) — Let U be a set of user stories, and let $US_i, US_j \in U$.

Let $A(US_i)$ and $A(US_j)$ denote the sets of activities associated with US_i and US_j, respectively. Let $R(US_i)$ and $R(US_j)$ denote the sets of reasons associated with US_i and US_j, respectively.

Each activity or reason $a \in A(US_i) \cup R(US_i)$ (and $b \in A(US_j) \cup R(US_j)$) is defined as a pair:

$$a = (a.\text{Action.name},\ a.\text{Entity.name}) \qquad b = (b.\text{Action.name},\ b.\text{Entity.name})$$

Define the set:

$$S := (A(US_i) \cup R(US_i)) \times (A(US_j) \cup R(US_j))$$

Define the following predicates:

- permits(a, b): One activity or reason establishes a necessary system service, condition, environment, configuration, or infrastructure such that other activity or reason can be meaningfully performed or deployed.

- $\text{OBD}(a, b)$: There exists an Operation-based Dependency between a and b (as previously defined).
- $a \approx b$: a and b are lexically identical or semantically equivalent, i.e., both pairs have `name`-attributes of both Action and Entity that are lexically identical or semantically equivalent; that is,

$$a.\text{Action.name} \approx b.\text{Action.name} \ \wedge \ a.\text{Entity.name} \approx b.\text{Entity.name}$$

A *Sequential Dependency* exists between US_i and US_j **if and only if**

$$\begin{aligned}
&\exists\,(a,b) \in S : \\
&\quad \big[(a,b) \in A(US_i) \times A(US_j) \ \ \wedge \ \ \neg\,\text{OBD}(a,b) \ \ \wedge \ \ \text{permits}(a,b)\big] \\
&\quad \oplus \\
&\quad \big[(a,b) \in \big(A(US_i) \times R(US_j) \cup R(US_i) \times A(US_j)\big) \ \ \wedge \ \ \big(a \approx b \ \vee \ \text{permits}(a,b)\big)\big] \\
&\quad \oplus \\
&\quad \big[(a,b) \in R(US_i) \times R(US_j) \ \ \wedge \ \ \text{permits}(a,b)\big]
\end{aligned}$$

In words: A *sequential dependency* between two user stories exists when an activity or reason in one story establishes the system conditions, such as services, configurations, or environments, that are necessary for an activity or reason in the other story to be meaningfully carried out. This relationship can occur whether the connection is based on actions, justifications, or both, and applies only if there is not already a more specific operation-based (entity enabling) dependency between them. In other words, sequential dependency describes cases where one story must be completed first to make it possible or logical to implement or perform the other.

Definition 2.8 (**Compositional Dependency - Pairwise**) — Let $E(US)$ be a set of entities which are introduced or manipulated by user story US.
A **Compositional Dependency** exists between US_i and US_j if and only if there exists a pair of entities (e_i, e_j) such that:

$$\begin{aligned}
e_i &\in E(US_i), \\
e_j &\in E(US_j), \\
\text{and} \quad \texttt{has-a}(e_i, e_j)&.
\end{aligned}$$

where $\texttt{has-a}(e_i, e_j)$ denotes denotes Meronymy-Holonymy (whole-part) relation, i.e., one of the entity (e_i or e_j) is a component, attribute, or substructure of another.

Definition 2.9 (**Hierarchical Dependency - Pairwise**) — Let $E(US)$ be a set of entities which are introduced or manipulated by user story US.
A **Hierarchical Dependency** exists between US_i and US_j if and only if there exists a pair of entities (e_i, e_j) such that:

$$\begin{aligned}
e_i &\in E(US_i), \\
e_j &\in E(US_j), \\
\text{and} \quad \texttt{is-a}(e_i, e_j)&.
\end{aligned}$$

where `is-a`(e_i, e_j) denotes Hyperonym- Hyponym (type-subtype) relation , i.e. one of the entity (e_i or e_j) is a subtype or specialization of another.

Definition 2.10 (**Self Reflect Dependency**) — Let U be a set of user stories, and let $US \in U$ be a user story.

Let $A(US)$ be the set of activities associated with US, where each activity $a \in A(US)$ is defined as a pair:

$$a = (a.\text{Action.name},\ a.\text{Entity.name})$$

Specifically, *Action* denotes a *GoalAction* and *Entity* denotes a *GoalEntity*.
Define the following:

- requires(a_i.Action.name, a_j.Action.name): The operation described in the `name`-attribute of Action of a_i strictly requires prior implementation of the operation described in the `name`-attribute of Action of a_j, *within the same user story.*
- a_i.Entity.name $\sim$ a_j.Entity.name: There exists a semantic relationship between the `name`-attributes of Entity of a_i and a_j, where $\sim$ indicates a form of semantic relatedness as relevant to the user story context **except** lexical identity and semantic equivalence(synonym).
- a_i.Entity.name $\approx$ a_j.Entity.name: Lexical identity or semantic equivalence between `name`-attributes of Entity of a_i and a_j.

A **Self-Reflect Dependency** exists in US **if and only if**

$$
\begin{aligned}
|A(US)| > 1 \ \wedge\ \exists\, a_i, a_j \in A(US)\ :\quad & \big[\, a_i.\text{Entity.name} \approx a_j.\text{Entity.name} \\
& \quad \wedge\ \ \text{requires}(a_i.\text{Action.name},\ a_j.\text{Action.name}) \,\big] \\
& \oplus \\
& \big[\, a_i.\text{Entity.name} \sim a_j.\text{Entity.name} \\
& \quad \wedge\ \ a_i.\text{Action.name} \approx a_j.\text{Action.name}) \,\big]
\end{aligned}
$$

In words: Self-Reflect Dependency captures internal depedencies within a single user story, where either the **actions** depend one each other and involve the **same entity** or the **entities** are semantically related but not lexically identical or sematically equivalent, while the **actions** are lexically identical or semantically equivalent.

3 Methodology

3.1 Dataset

Which quality criteria have been already checked in the paper of Hofmann?

3.2 PoC

3.3 Evaluation

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